SemiAI

SMILE: The AI Operating System for Semiconductor Yield

Predictive control, multimodal root-cause inference, and virtual fab simulation for advanced chip manufacturing.

Pre-A // U.S. EXPANSION

Taekwon Jee // CEO & Founder
SemiAI

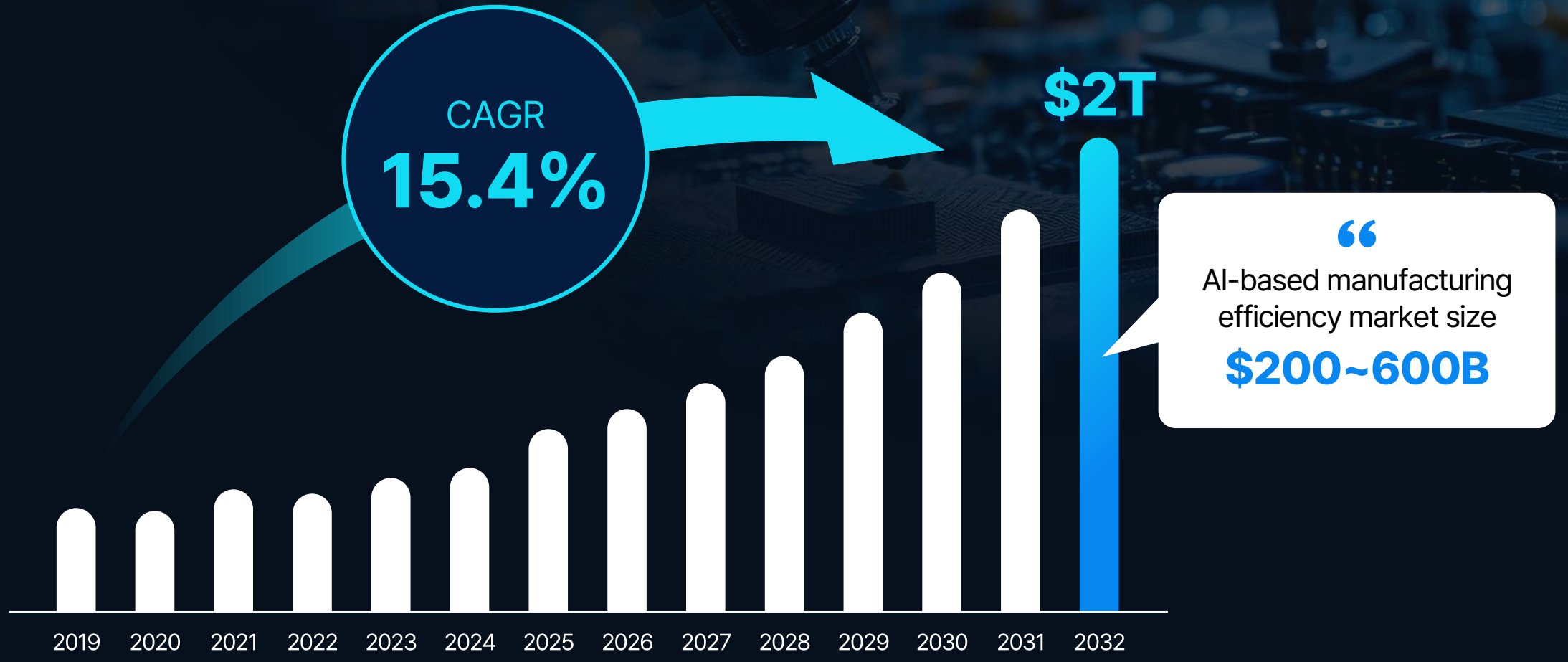
Why it resonates now

91%	0.3nm	R^2>0.9
Top-5 causal retrieval	Prediction accuracy	Hotspot prediction



Validated alpha KPI signals yield upside while reducing engineer iteration and fab waste.

Rapid growth in the semiconductor market with AI

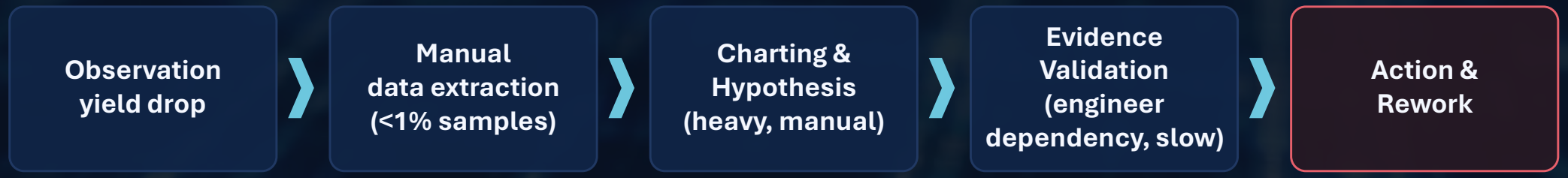


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AI-based manufacturing efficiency market size
$200~600B

Sorce : Fortune BI, May 2025

Fabs still fight yield losses with slow, reactive workflows

Today’s root-cause analysis loop is chart-heavy, engineer-dependent, and too slow for advanced-node complexity.



Delay
Late detection

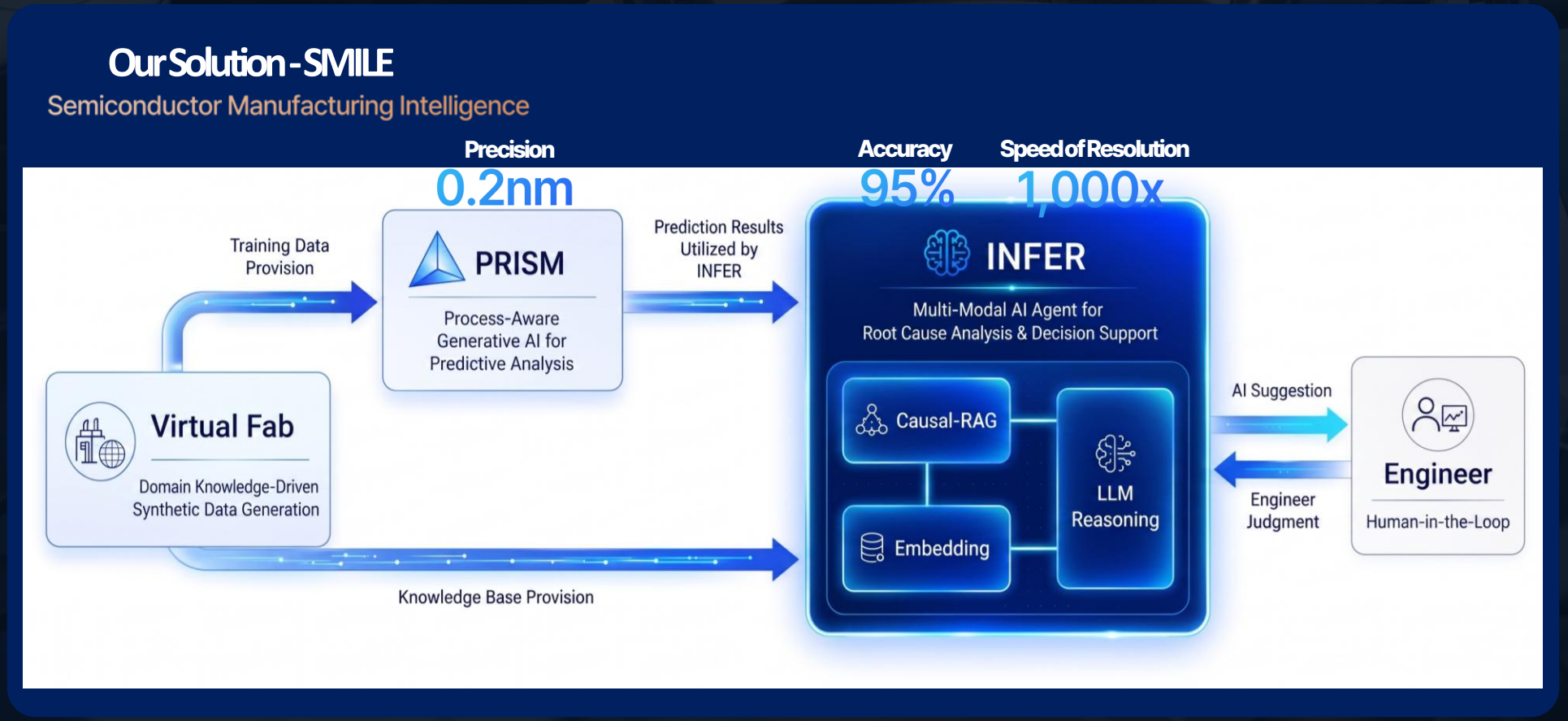
Complexity
Advanced-node workflows

Waste
Yield and engineering drag

→ Fabs don’t need more dashboards. They need an AI decision loop.
Predict earlier. Explain faster. Act before yield loss grows.

SMILE closes the loop from signal to decision

A fab-native AI platform that combines prediction, multimodal inference, and virtual simulation with engineers in the loop.



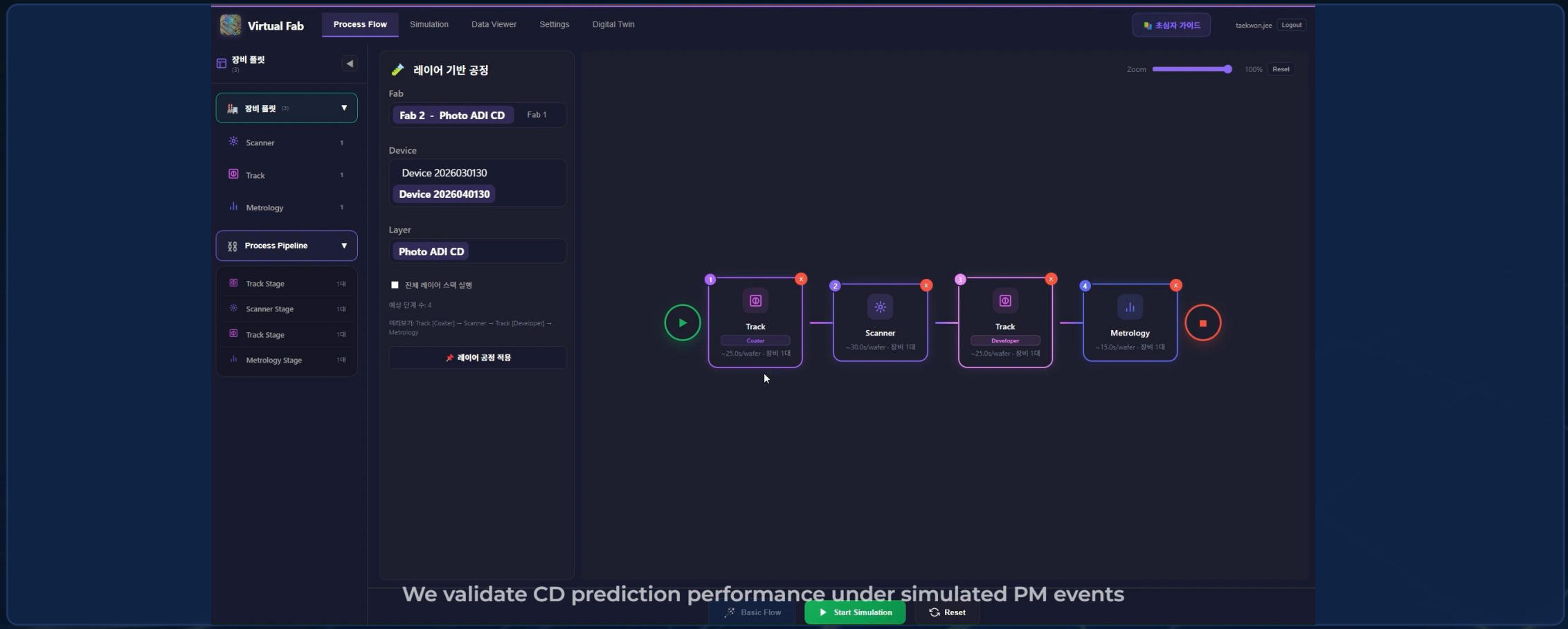
Speed-Up
Early detection

Efficiency
Agentic AI workflows

Cost Saving
High yield, less rework

Virtual Fab + PRISM: Predicting CD from FDC signals

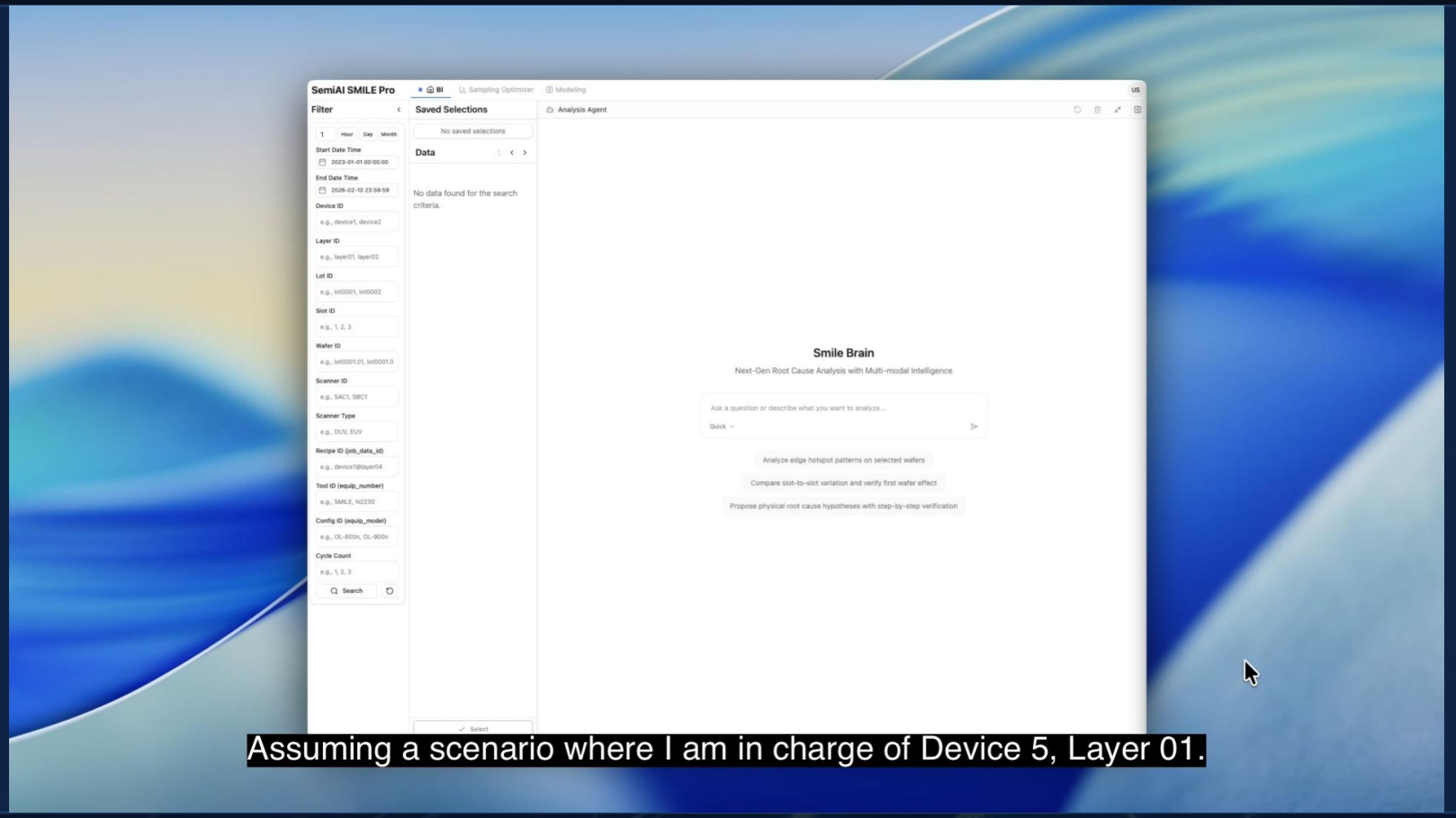
AI-powered virtual metrology for faster process feedback and earlier action (https://youtu.be/jMAj8SV_WVU)



Synthetic Data | Process Simulation | Scenario Generation | Predictive Learning

Agentic AI: From fragmented data to evidence-backed root cause analysis

Multimodal reasoning across charts, reports, history, and fab signals (https://www.youtube.com/watch?v=B63cKxStWkw)



AI Chart Analysis | AI-generated Report | Conversational RCA | Code + Stats + Simulation

Land-and-expand business model built for global fabs

Start with a focused pilot, prove yield or cycle-time value, then expand from one line to fab-wide deployment.

SMILE Lite

HW+SW bundle

- Dashboards + correlation
- Sampling / metrology efficiency

→ Copy license per Hardware

SMILE Pro

Fab analytics / one use case

- Virtual Fab simulation
- Process optimization for fabs + OEMs

→ Annual subscription for mid-size fabs (one line)

SMILE Pro+

Premium contracts / Killer options / multiple use cases

- PRISM predictive intelligence
- INFER + Semi-Brain copilot

→ Annual subscription for Tier-1 fabs (one line)

Target buyers

Chip-makers

FEOL / BEOL

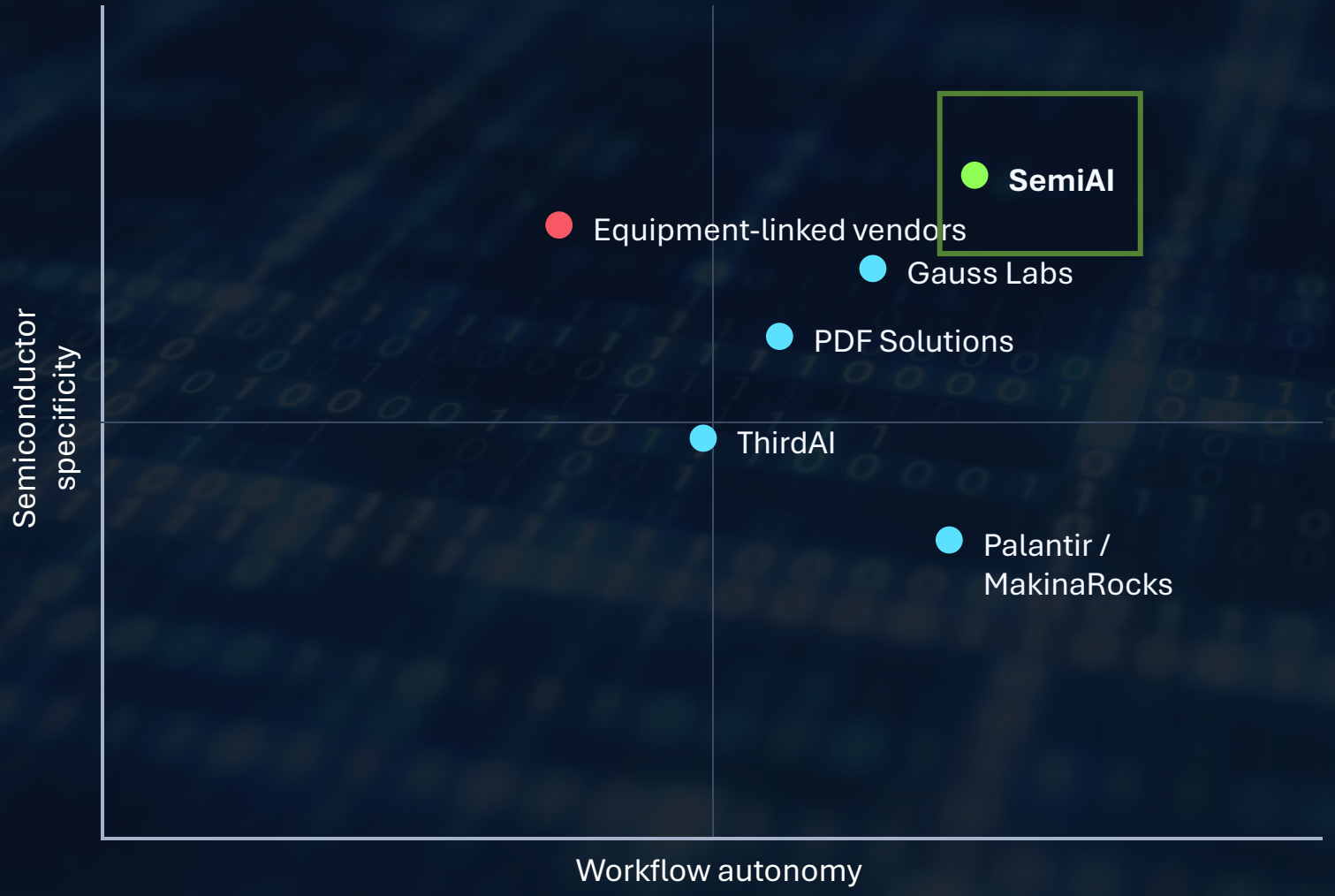
Equipment OEMs

Advanced packaging

Recurring software + expansion modules create asymmetric upside as customer data, workflows, and trust compound.

Most integrated semiconductor-native stack

White space: deep fab knowledge + autonomous workflow.



Why we win

Prediction + reasoning + automation
Day-1 usability
Rare-event robustness from synthetic fab data

Most relevant rivals: Gauss Labs, PDF Solutions, ThirdAI

Commercial traction is forming before U.S. expansion

SemiAI already has early customer validation, strategic visibility, and a growing IP base — the next step is local customer acquisition.

3
VCs for Seed Funding
Including Kakao

5
Multiple POCs
in parallel

$33K+
Early Revenue
POC revenue in Year 1.5

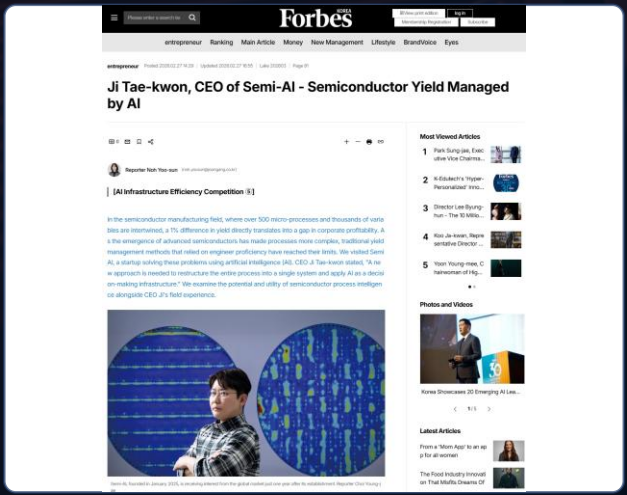
5
Patent Filings
in 12 months

SAMSUNG · SK hynix · DB HiTek · Magnachip

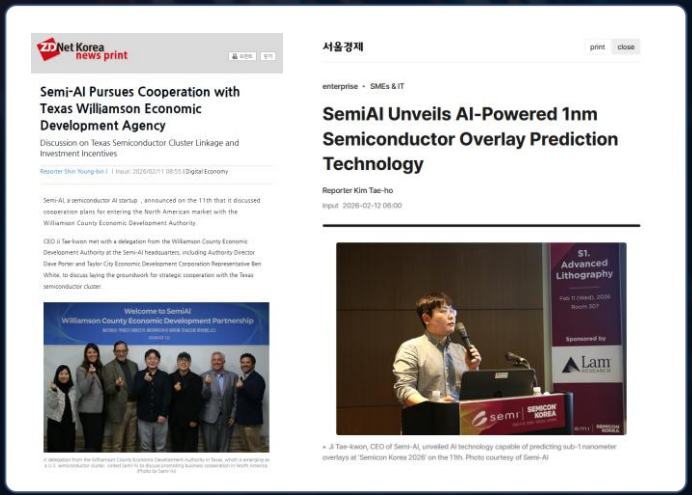
Strategic validation

NVIDIA · Applied Materials · Lam Research · TEL

Texas LOI: Williamson County EDA


Forbes

Ji Tae-kwon, CEO of Semi-AI - Semiconductor Yield Managed by AI


Net Korea news print

Semi-AI Pursues Cooperation with Texas Williamson Economic Development Agency

서울경제

SemiAI Unveils AI-Powered 1nm Semiconductor Overlay Prediction Technology

External visibility helps de-risk customer acquisition and hiring in the U.S. market.

- Forbes feature
- 1nm overlay prediction launch
- Conference ecosystem visibility

A founder-market fit team for semiconductor + AI execution



Taekwon Jee

CEO & Founder

- PhD, UC Berkeley
- 15+ yrs at SK hynix, ASML, Samsung, Lam Research
- Built AI systems for yield improvement
- Deep process-control / EPE expertise

UC Berkeley · SK hynix · Lam Research · SAMSUNG · ASML



Jenghun Suh

AI Research Lead

- PhD in EECS, SNU
- +7yrs in AI system (Samsung Electronics)

Seoul National University · SAMSUNG



Jihoon Jung

Product Design Lead

- MS in EECS, KNU
- +10yrs in Semi HW (KLA & ASML)

KNU Kyungpook National University · ASML · KLA



Jungsuk Kim

Advanced Tech Lead

- PhD in CE, Purdue
- +6yrs in Semi Mat. (Dupont)

Purdue University · DUPONT



Taekeun Kim

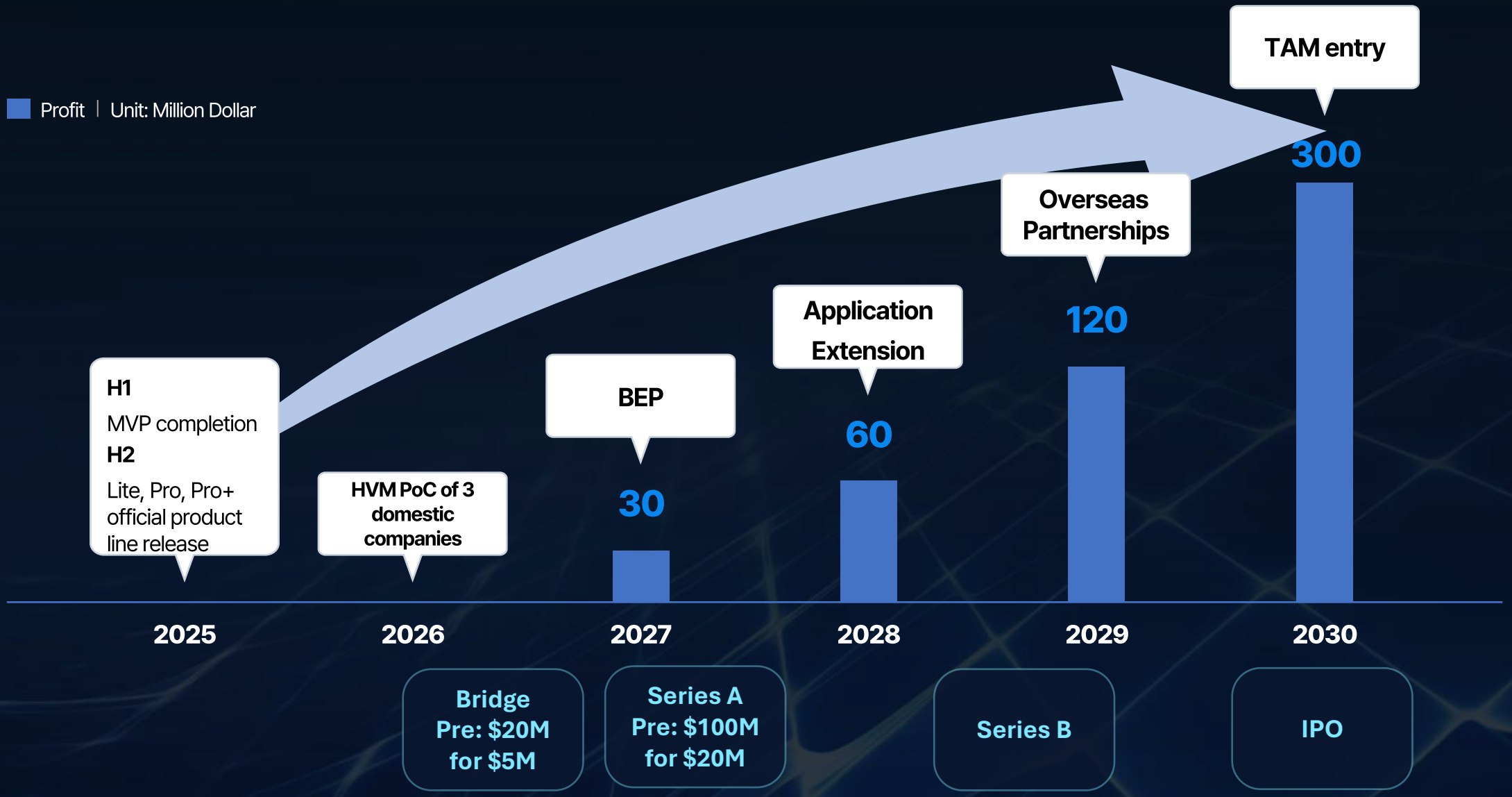
Sales/Marketing Dir.

- MBA, SUNY
- +20yrs in Marketing. (Hyundai Motors, Hanhwa etc.)

State University of New York · HYUNDAI · Hanwha

- **Operating DNA: deep semiconductor process expertise + software + AI research**
- **Backed by a broader team from leading institutions and industry leaders.**

Roadmap and Sales plan



Raising $5M Pre-A to win the first U.S./EU Tier 1 fab customer

We are using this round to translate model proof into local commercial execution and premium recurring contracts.

$5M

Pre-A round

Use of funds

- Local U.S./EU applications and technical sales
- Compute and R&D scaling for SMILE deployment
- Customer pilots, onboarding, and fab integration support

Bridge to Series A

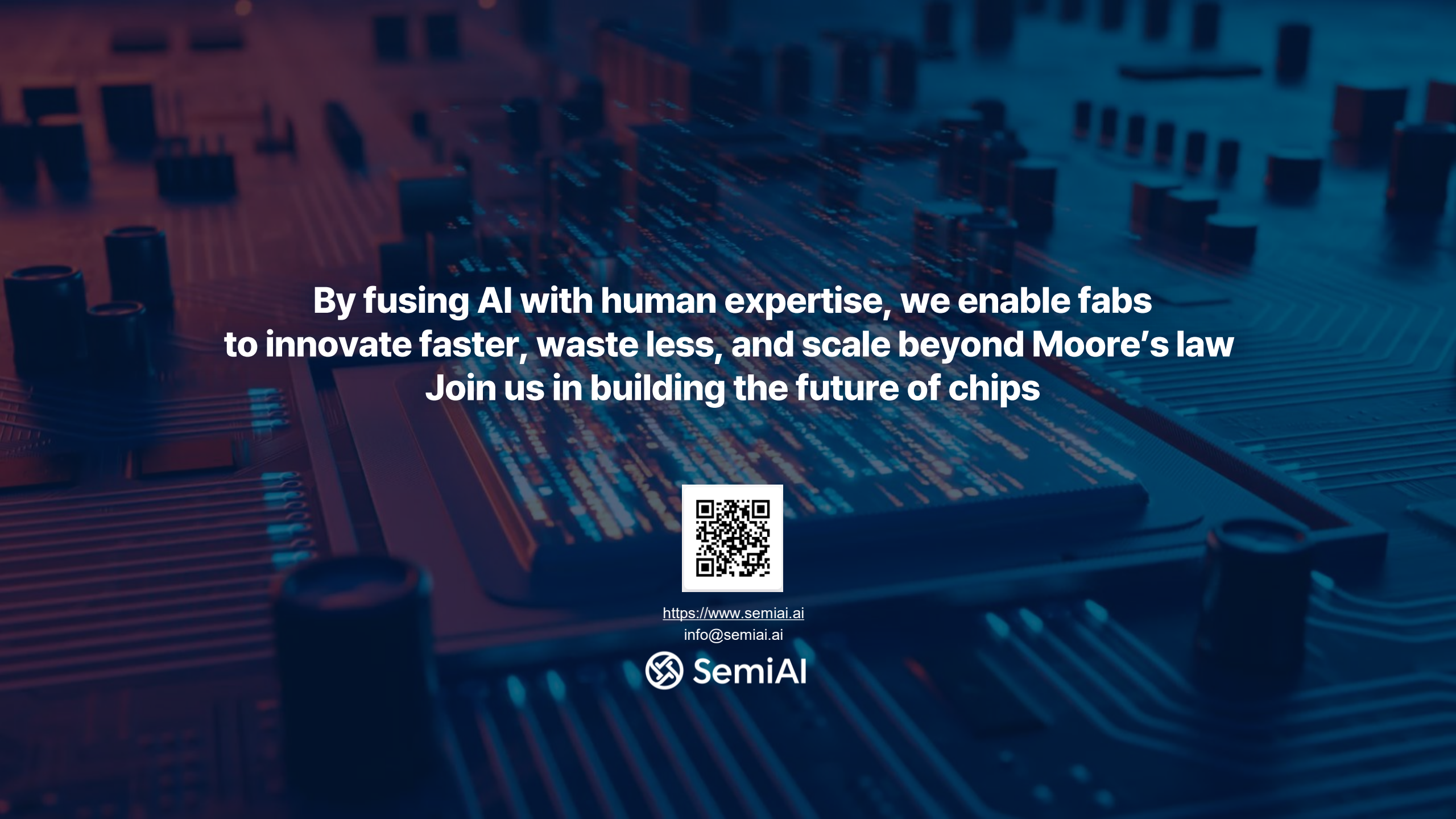
1st U.S./EU customer
3–5 active POCs
~$500K ARR
Tier 1 revenue quality

Why invest now

- A rare combination of fab-domain credibility, proprietary synthetic-data generation, and multimodal AI inference.
- Clear early signals on product performance, commercial traction, and strategic ecosystem interest.
- U.S. expansion can convert those signals into durable, high-value software revenue with Tier 1 fabs.

By fusing AI with human expertise, SemiAI helps fabs innovate faster, waste less, and scale beyond Moore's law.

Let's build the future operating layer for semiconductor manufacturing.


By fusing AI with human expertise, we enable fabs
to innovate faster, waste less, and scale beyond Moore's law
Join us in building the future of chips
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